

FACULTY OF BUSINESS AND IT

MBAI 5600G - Applied Integrative Analytics Capstone Project

Milestone 6 Report

**AI-Based Student Dropout Early Warning System for Higher Education:
Validation and Business Impact Assessment on UCI and OULAD**

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1. Executive Summary

Milestone 6 validates the two final early-warning models from Milestones 4 and 5. We did not add another algorithm; the goal was to check whether the selected models are reliable enough for a small pilot. The UCI model is a tuned Logistic Regression model for structured student-record data. The OULAD model is a tuned XGBoost model for a larger online-learning dataset with assessment and platform-activity signals. On the held-out test set, UCI reached F1 = 0.793, recall = 0.820, and ROC AUC = 0.911. OULAD reached F1 = 0.738, recall = 0.719, and ROC AUC = 0.885. We also tested repeated stratified splits, 10% simulated missingness, 5% numeric noise, error patterns, and subgroup performance. Numeric noise had little effect. Missing values reduced performance more, especially for UCI. Per 1,000 students, the UCI model would flag about 344 students and correctly identify about 263 at-risk students. The OULAD model would flag about 295 and correctly identify about 224. The recommendation is a limited advisor-support pilot, with monitoring of missing data, subgroup recall, and alert workload.

2. Model Validation

We validated the models using the same target labels and feature windows used in the previous milestones. This choice matters because changing the setup at this stage would make the Milestone 5 results hard to compare.

The UCI early-warning table still removes second-semester fields. The OULAD table still uses the first 75 days of course activity. These settings are consistent with the leakage-control approach from Milestone 4. After tuning, each final model was trained on the combined training and validation data. The final evaluation was then done on the previously unused 20% test set.

Final model	Rows	Test rows	Cutoff	Precision	Recall	F1	ROC AUC
UCI Logistic Regression	4,424	885	0.50	0.766	0.820	0.793	0.911
OULAD XGBoost	32,593	6,519	0.60	0.758	0.719	0.738	0.885

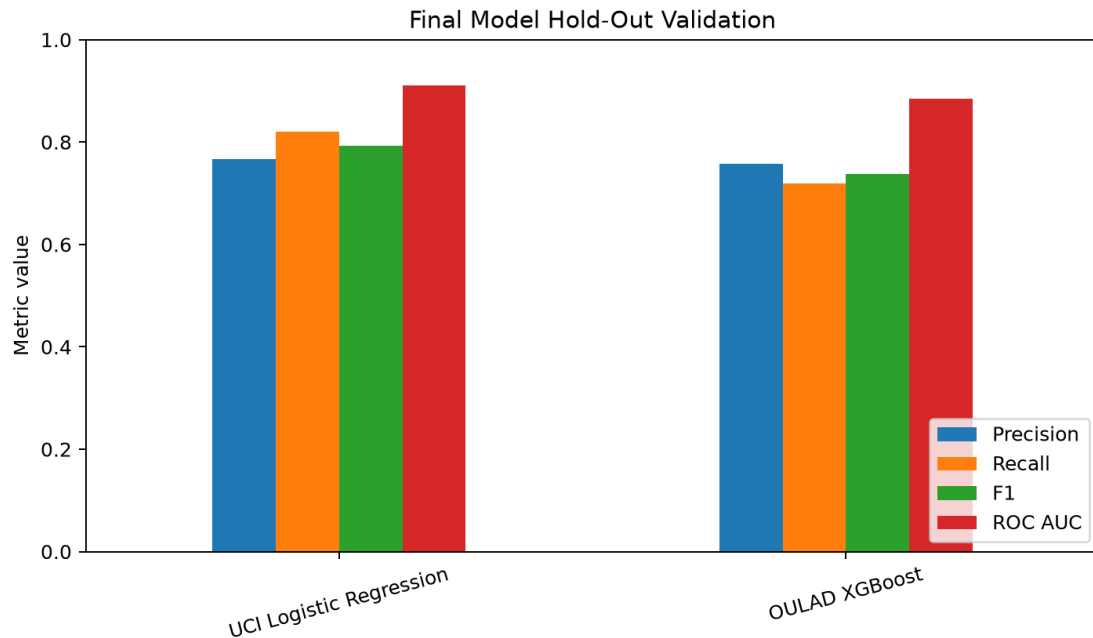


Figure 1. Hold-out validation metrics for the final early-warning models.

As expected, the UCI model performed better on most major metrics. This is mainly because the UCI data is cleaner and more directly connected to first-semester academic progress. The OULAD model has a harder task because online behavior data is noisier and varies by course. Even so, it still adds value because it reflects the type of information an online course team could monitor while a course is still active.

3. Robustness and Reliability Assessment

To check robustness, we used two practical tests. First, we reran each final model with five different stratified train-test splits. This helped us see whether the result depended too much on one random split. Second, we stress-tested the held-out test set with 10% simulated missing values and 5% numeric noise. These tests are simple, but they reflect issues that can happen in real student data.

Model	Splits	F1 mean/std	Recall mean/std	ROC AUC mean/std
OULAD XGBoost	5	0.738 +/- 0.005	0.722 +/- 0.011	0.884 +/- 0.004
UCI Logistic Regression	5	0.776 +/- 0.023	0.797 +/- 0.044	0.904 +/- 0.016

Model	Scenario	F1	Recall	Changed pred.	Score corr.
UCI Logistic Regression	Original	0.793	0.820	0.0%	1.000
UCI Logistic Regression	10% missing	0.753	0.746	5.3%	0.935
UCI Logistic Regression	5% numeric noise	0.788	0.813	0.5%	0.999
OULAD XGBoost	Original	0.738	0.719	0.0%	1.000
OULAD XGBoost	10% missing	0.718	0.673	3.4%	0.973
OULAD XGBoost	5% numeric noise	0.734	0.718	2.3%	0.987

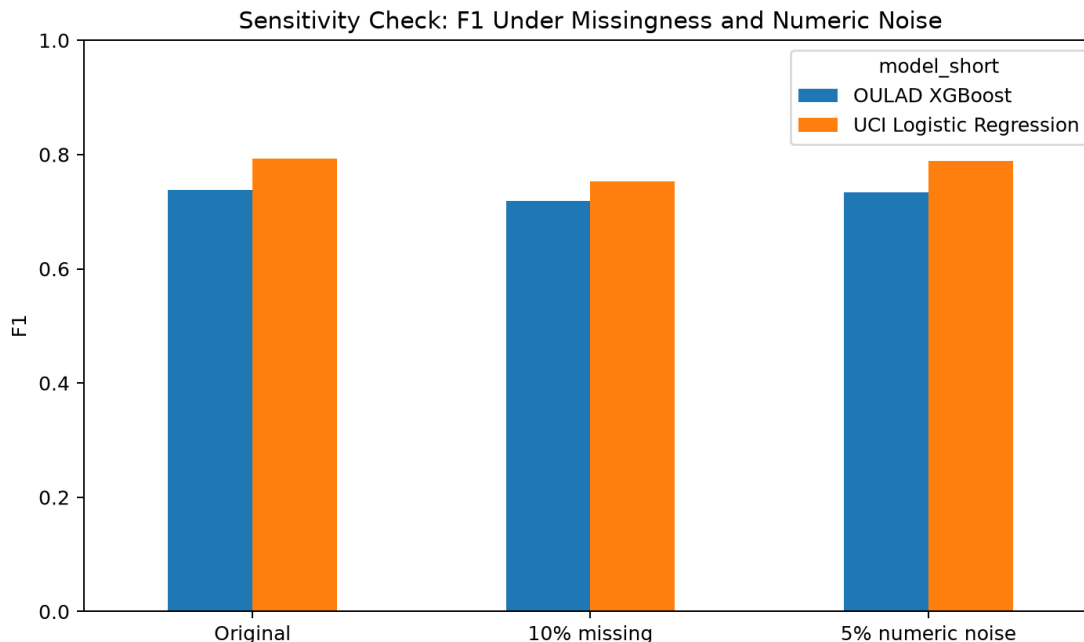


Figure 2. F1 sensitivity under missingness and numeric-noise stress tests.

Numeric noise had little effect on the test results. Missing data caused a larger drop. UCI F1 decreased from 0.793 to 0.753, and OULAD F1 decreased from 0.738 to 0.718. This does not make the models

unusable, but it does show that data completeness should be checked before scores are shown to advisors.

4. Error Analysis

Model	TP	FP	FN	TN	Precision	Recall
UCI Logistic Regression	233	71	51	530	0.766	0.820
OULAD XGBoost	1460	466	571	4022	0.758	0.719

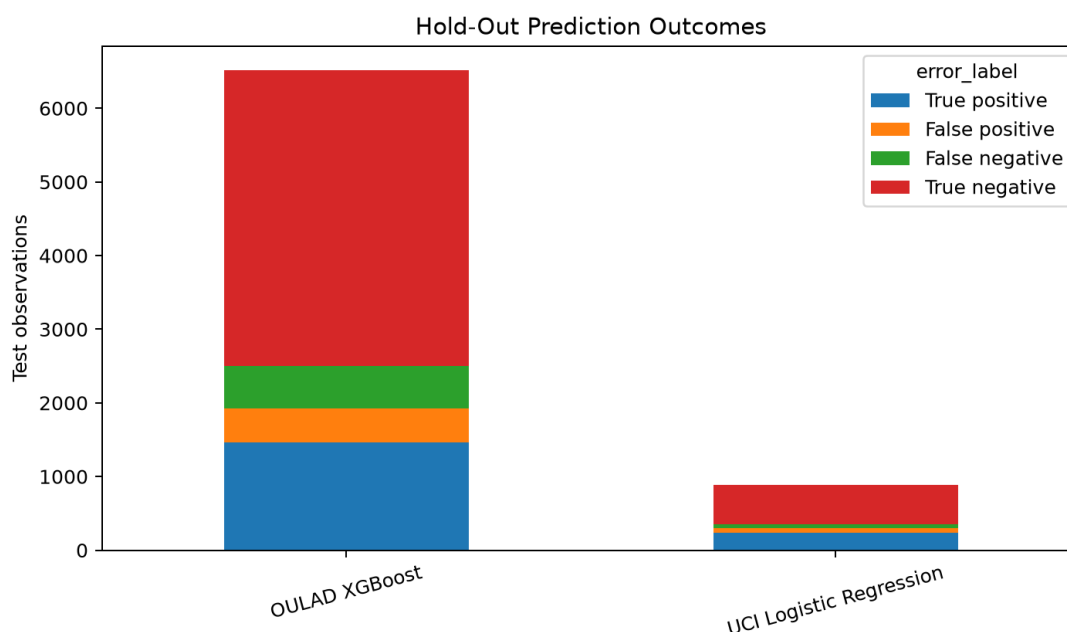


Figure 3. Hold-out prediction outcomes for the final models.

For an early-warning system, the most important error type is the false negative. A false negative means the student is actually at risk but does not appear on the advisor list.

The UCI test set had 51 false negatives, while the OULAD test set had 571. This difference is partly due to sample size, because the OULAD test set is much larger. It is also related to the selected threshold. OULAD uses a more conservative cutoff, so some at-risk students fall below the alert threshold.

The feature patterns help explain these errors. In UCI, true positives usually had weak first-semester academic progress. False negatives looked more like continuing students when measured by approved units and grades. In OULAD, true positives usually had low early assessment activity and low platform use. False negatives showed more engagement, including course activity and online participation, which made them harder to separate from students who stayed enrolled.

4.1 Error Feature Profile

Dataset	Feature	TP mean	FN mean
UCI	Curricular units 1st sem (approved)	1.867	6.078
UCI	Curricular units 1st sem (grade)	6.025	12.278
UCI	Curricular units 1st sem (enrolled)	5.639	7.118
OULAD	Early assessment submission ratio	0.131	0.885
OULAD	Highest early assessment score	15.045	76.877
OULAD	Early weighted-score ratio	0.053	0.584

5. Business Impact Assessment

The business value of this model depends on whether it can produce an advisor list that staff will actually use. F1 is useful, but it does not tell the whole story. A model with too many alerts could overwhelm advisors, while a model with too few alerts could miss students who need help. For that reason, the results were converted into alerts per 1,000 students. The dollar values in this section are planning assumptions, not confirmed institutional financial estimates.

Model	Alerts/1,000	True at-risk found	False alerts	Missed at-risk	Advisor weeks
UCI Logistic Regression	343.5	263.3	80.2	57.6	13.7
OULAD XGBoost	295.4	224.0	71.5	87.6	11.8

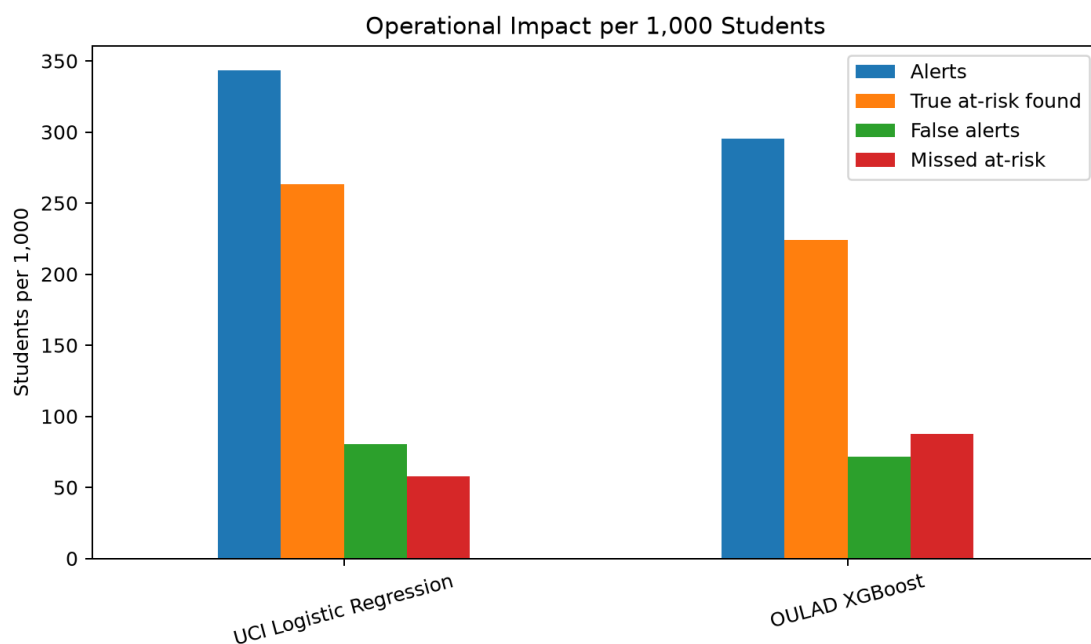


Figure 4. Operational impact estimates per 1,000 students.

Using a rough outreach cost of \$100 per alert and a \$6,000 value for retaining one additional student, the break-even point is small. The institution would need to retain about 5 to 6 additional students per 1,000 students. This is not a guaranteed return on investment, because we do not yet know how many flagged students would respond to outreach. It does suggest that a controlled pilot is reasonable.

6. Business Interpretation

The model should serve as an advising aid, not as a final evaluation of a student. A high-risk score does not mean that the student will fail or withdraw. It means the student's record looks similar to past students who needed support. The advisor would still decide whether academic advising, financial counseling, course-engagement support, or a simple check-in is the right response.

UCI and OULAD support different use cases. UCI is better suited to a traditional student-record system where first-semester grades, approved units, and financial-status fields are available. OULAD is better suited to an online-learning setting where assessment submissions, scores, and platform activity can be reviewed while the course is still running.

The main benefit is timing. Without an early-warning model, many at-risk students may only be noticed after they fail or withdraw. With the model, support staff can review risk earlier and offer help before the situation becomes worse. This approach only works if the risk score is connected to a real advising process and if the alert volume remains manageable.

7. Deployment Feasibility Assessment

Area	Assessment	Rationale
Data pipeline	Feasible	UCI-style deployment requires updated first-semester records. OULAD-style deployment requires early assessment records and learning-platform aggregation.
Infrastructure	Low to moderate	Both models train quickly on the current datasets. A scheduled batch job should be enough for a small pilot.
Scalability	Feasible	OULAD contains more than 32,000 records and still runs locally, so institutional batch scoring appears feasible.
Integration	Moderate	The main challenge is connecting model output to advising workflows and deciding which advisors receive alerts.
Monitoring	Required	Each term, track missing data, alert volume, precision, recall, subgroup performance, and advisor feedback.
Update strategy	Term-based	Retrain or recalibrate after each academic term, especially if course structures or student populations change.

A full automated rollout would be unsafe at this stage. A better next step is a small pilot, such as a weekly advisor list or dashboard for a few programs or online modules. The pilot should answer questions that metrics alone cannot answer. Do advisors use the alerts? Is the list too large? Are some students missing because scores are unavailable? Does outreach change student outcomes? If the pilot results are positive, the system can be expanded slowly and reviewed after each term.

8. Ethical, Legal, and Risk Assessment

The main ethical concern is that an alert could become a label. A student should not be treated as a problem case only because of a risk score. The model also includes demographic and socio-economic proxy variables, so subgroup monitoring remains necessary even when the goal is student support.

The table below summarizes the main risk areas and safeguards for a pilot deployment.

Fairness should be reviewed by subgroup, including gender, disability, scholarship status, tuition status, and educational background. Advisors should also see the main reason categories behind alerts, such as academic progress, engagement, and financial-status indicators. Alerts should lead to support, not penalties or reduced opportunities.

For legal and policy reasons, the model should be used in a narrow way. Scores should support student-success triage only. Approved uses should be documented, access should be limited to staff who need the information, and no negative decision should be made solely from model output.

Risk area	Mitigation
Privacy	Use only the data needed for support decisions, restrict access to risk outputs, and avoid exposing raw sensitive fields to unnecessary users.
Fairness	Review performance by available groups such as gender, disability, scholarship status, tuition status, and educational background.

Transparency	Show advisors the top reason categories behind alerts, especially academic progress, engagement, and financial-status signals.
Unintended consequences	Prevent punitive use. Alerts should trigger support, not automatic exclusion, grading consequences, or reduced opportunity.
Security	Store outputs in approved institutional systems and audit who can view student-level risk scores.

9. Final Conclusions and Recommendations

Overall, the validation results support a cautious pilot. The models still work on held-out data and remain reasonably stable across repeated train-test splits. Their error patterns can also be explained. However, missing data and false negatives remain important risks. For this reason, the system should remain human-in-the-loop and should not be used as an automated decision tool.

- Use the first pilot as advisor decision support only, not as automated decision-making.
- Use the UCI Logistic Regression model when the main data source is structured student records.
- Use the OULAD XGBoost model when early assessment and online-learning activity data are available.
- After each term, review alert volume, missing data, subgroup recall, and actual support outcomes.
- Future work should test other OULAD time windows, calibration methods, and intervention-response data.

10. References

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